

Skills

Languages: Python, TypeScript, Bash, Solidity

Software Engineering: Git, Linux, GitHub Actions, CI/CD, pytest, Python packaging, Ruff, mypy, Next.js

Quantum / Numerical: Qiskit, PennyLane, QEC tooling, VQE, tensor networks, NumPy, SciPy

Experience

2023–2026 **Quantum Software Lead**, *BTQ*, Vancouver, Canada

- Led planning and work allocation for a **four-person quantum software sub-team**, coordinating multiple concurrent deliverables and research integration.
- Developed **QLDPC**, a modular Python QEC toolkit combining interactive circuit construction, belief-propagation decoding workflows, Qiskit integration, and code/decoder visualization.
- Designed **Léonne** for modular consensus and topological trust partitioning, and built **QRiNG**'s Python simulation/validation layer around an existing Solidity contract for verifiable quantum randomness.
- Led BasiQ and Qonductor development in TypeScript/Next.js, defining domain content, application behavior, UI/UX, and external partner integration; maintained QByte backend metrics.

2024–2026 **Quantum Algorithms Researcher**, *University of Victoria*, Victoria, Canada

- Built Python/Julia exact-diagonalization and tensor-network workflows across quantum many-body models, comparing integrable and non-integrable dynamics through level statistics, observables, and time evolution.
- Quantified relaxation and equilibration scaling across system size, coupling regimes, and initial states, using exact and tensor-network methods to characterize coherence-relevant timescales.
- Coordinated **two junior researchers**, assigning computational tasks and mentoring one student while integrating results into the main analysis workflow.

Earlier Experience

2023–2024 **Quantum Machine Learning Researcher**, *Fudan University*, Shanghai, China

- Built **TQNN**, a pip-installable Python toolkit with automated tests, CI, command-line entry points, and three interactive tensor-network simulation/classification tools.
- Implemented gradient-free classification with spin-network/TQFT-inspired encodings and evaluated robustness under increasing noise.

2025 **Quantum Computing Researcher**, *UBC / Institut Quantique*

- Implemented an **H₂ variational quantum eigensolver** from Jordan–Wigner mapping and Pauli-operator construction through parameterized circuits and exact-diagonalization validation, then packaged it with CLI entry points, automated tests, linting, and type checking; awarded 2nd place for visualization and presentation.

2019–2020 **Quantum Simulation Developer**, *Vanier College*, Montréal, Canada

- Solved non-linear Hamilton–Jacobi PDEs with Crank–Nicolson methods and implemented RNN-driven quantum-trajectory simulations for arbitrary potential landscapes in Python. 

Education

2026 **M.Sc. Physics**, *University of Victoria*, Victoria, Canada

- Thesis: *Scaling Analysis and Ensemble Measures in Quantum Systems*; scholarships: NSERC CREATE Quantum Computing Program, BCGS, UVic FGS (\$31,500).

2023 **B.Sc. Honours Physics**, *McGill University*, Montréal, Canada

- Honours thesis: *Conflicts with de Sitter vacua in supersymmetric field theories*; published as *Conflicts with de Sitter Vacua in Superstring Theory*. Scholarships: 2 NSERC USRAs + FRQNT, SURA, SURE, BSA (\$41,600).